

Training GNNs					
Mode	Configuration	# Parameters	Time (Seconds)		
			HuggingFace	Multimedia	Daily Life
GNN-only	GCN	1,049,600	136.5	136.1	237.7
	GAT	1,051,648	136.9	151.8	237.8
	GraphSAGE	2,098,176	134.2	149.8	233.8
	GIN	2,099,200	134.2	134.9	233.6
	GraphTransformer	4,198,400	135.0	150.2	233.7
LM+GNN Co-trained	LM+SGC	335,141,889	743.1	323.3	384.7
	LM+GCN	336,191,488	482.8	567.8	384.2
	LM+GAT	336,193,536	741.3	812.3	384.4
	LM+GraphSAGE	337,240,064	741.2	406.8	381.7
	LM+GIN	337,241,088	735.6	361.8	382.9
	LM+GraphTransformer	339,340,288	741.0	362.7	405.8
Fine-tuning LLMs					
LLM	# Param (# Trainable Param)	Device & Time (Hours)			
		HuggingFace	Multimedia	Daily Life	
CodeLlama-7B	6,742,740,992 (4,194,304)	2×A100 7.0	1×A100 13.8	2×A100 9.5	
Vicuna-13B	13,022,417,920 (6,553,600)	1×A100 17.8	2×A100 10.3	2×A100 19.0	